

Xueren Ge

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BACKGROUND

University of Virginia

Ph.D. in Computer Engineering GPA: 3.88/4.0

Jun. 2022 – Present

Charlottesville, Virginia

Georgia Institute of Technology

M.S. in Electrical and Computer Engineering GPA: 3.91/4.0

Aug. 2020 – May. 2022

Atlanta, Georgia

Chongqing University

B.E. in Electrical Engineering and Automation GPA: 3.58/4.0

Sep. 2016 – Jun. 2020

Chongqing, China

COURSEWORK

Statistical Machine Learning, Natural Language Processing, Deep Learning, Computer Vision, Reinforcement Learning, Convex Optimization, Random Processes, Computer Communication Networks, Intro to Database System, Object-Oriented Programming, Data Structures & Algorithms, Embedded Systems, Learning for Interactive Robots, Discrete Math, Linear Algebra, Calculus, Probability & Statistics

SCHOLARSHIPS & AWARDS:

- AAI-26 Student Scholarship and Volunteer Program, 2026.
- Georgia Tech Shenzhen Campus *Merit-Based Scholarship (Level A)*, 2020 (top 5%).
- Yangtze Power Scholarship, 2018 (2/324, Top 0.01%).
- Excellent Second Undergraduate Comprehensive Scholarship, 2016/2018/2019 (top 6%).
- Chongqing University Excellent Undergraduate, 2019 (Top 10%)
- Chongqing University Excellent Student, 2018 (Top 5%)
- Chongqing University Science and Technology Innovation Advanced Individual, 2018 (Top 5%)
- Mathematical Contest in Modeling (MCM) — *Meritorious Winner*, 2018 (Top 10%)

RESEARCH INTERESTS

Natural Language Processing
Knowledge-Augmented LLMs
Multimodal Learning

Large Language Models
Uncertainty Estimation
Computer Vision

Conversational Recommendation Systems
Synthetic Data Generation
Activity Recognition

SKILLS

Languages: C/C++, Python, MySQL, Java, JavaScript/TypeScript, HTML/CSS, MATLAB, $\text{\LaTeX}$

Tools: Slurm, Git, Linux, Docker, Microsoft Azure, AWS, HPC Systems, GitHub, Unix Shell, VSCode

Machine Learning Frameworks: PyTorch, Tensorflow, Wandb, Huggingface, DeepSpeed, FSDP, DDP, vLLM

Embedded System: STM32, Keil, Multisim, Proteus, AutoCAD

PUBLICATIONS

- **Ge, X.**, Murtaza, S., Cortez, A., & Alemzadeh, H. (2026). *EMSDialog: Synthetic Multi-person Emergency Medical Service Dialogue Generation from Electronic Patient Care Reports via Multi-LLM Agents*. In **ACL Findings 2026**. (Acceptance rate: 37%).
- **Ge, X.**, Murtaza, S., Cortez, A., & Alemzadeh, H. (2025). *Expert-Guided Prompting and Retrieval-Augmented Generation for Emergency Medical Service Question Answering*. In **AAAI 2026, Main Technical Track**. (**Oral**, Acceptance rate: 17.6%).
- Weerasinghe, K., **Ge, X.**, Heick, T., Wijayasingha, L. N., Cortez, A., Satpathy, A., Stankovic, J. A., & Alemzadeh, H. (2025). *EgoEMS: A High-Fidelity Multimodal Egocentric Dataset for Cognitive Assistance in Emergency Medical Services*. In **AAAI 2026, Social Impact Track**. (Acceptance rate: 24.1%).

- Li, Z., Luo, X., **Ge, X.**, Zhou, L., Lin, X., & Liu, Y. (2025). *MMSense: Adapting Vision-based Foundation Model for Multi-task Multi-modal Wireless Sensing*. In **IEEE ICC 2026**.
- **Ge, X.**, Satpathy, A., Williams, R. D., Stankovic, J. A., & Alemzadeh, H. (2024). *DKEC: Domain Knowledge Enhanced Multi-Label Classification for Diagnosis Prediction*. In **EMNLP Main 2024**. (Acceptance rate: 18.4%.)
- Weerasinghe, K., Janapati, S., **Ge, X.**, Kim, S., Iyer, S., Stankovic, J. A., & Alemzadeh, H. (2024). *Real-Time Multimodal Cognitive Assistant for Emergency Medical Services*. In **IoTDI 2024**. (Acceptance rate: 36.7%.)
- Fang, X., Wang, B., Kong, H., **Ge, X.**, Yang, Z., Yu, J., & Li, W. (2023). *Human posture feature recognition method for neuropsychological comprehension test*. In **Journal of Chongqing University**.
- Ma, Y., & **Ge, X.** (2019). *An Effective Method for Defect Detection of Copper Coated Iron Wire Based on Machine Vision*. In **IOP Conference Series: Materials Science and Engineering**.

PATENTS

- Yu, H., Wang, B., Kong, H., He, W., **Ge, X.**, Yang, W., Li, W., Yang, Z., Lü, Y. (2025). *An action recognition method based on TensorFlow target detection*. **China Patent CN111860103B** (granted Jul. 15, 2025; assignee: Chongqing Zhiyixing Technology Development Co., Ltd.).
- Yu, H., Kong, H., **Ge, X.**, Wang, B., Wang, Z., Li, W., Yang, Z., Yu, W. (2024). *Automatic gesture recognition system for AD (analog-to-digital) meter understanding capability test*. **China Patent CN111652076B** (granted; assignee: Chongqing Zhiyixing Technology Development Co., Ltd.).
- Li, R., **Ge, X.**, Li, Q., Zhao, M., Bao, M., Ma, J. (2021–2024). *Respiratory rate detection method and device, electronic equipment, and storage medium*. **China Patent CN113887474B** (granted; assignee: Shenzhen Sensetime Technology Co., Ltd.).
- Li, R., **Ge, X.**, Li, Q., Chen, C., Ma, J. (2023). *Heart rate measurement method and apparatus, and electronic device and storage medium*. **WIPO Patent WO2023061042A1** (published Apr. 20, 2023; assignee: Shanghai Sensetime Intelligent Technology Co., Ltd.).
- **Ge, X.** (2019). *Highway electronic information direction board*. **China Patent CN209118652U** (granted Jul. 16, 2019; assignee: Individual).

INTERNSHIP EXPERIENCE

Samsung Research America

May. 2026 – Aug. 2026

LLM/Gen AI Research Intern | Mentor: Jilong Kuang

Mountain View, CA

• Self-improving Multi-Agent System for FHIR Question Answering

Agents, Skills, Self-evolving, Tool-using, Information Retrieval

- * Developed a Planner–Specialist–Composer multi-agent pipeline for question answering over FHIR patient bundles, decomposing complex queries into resource-specific subtasks (Observation, Medication, Encounter, Condition, Procedure), routing to domain specialists that perform LLM-guided field selection and ReAct-style iterative retrieval reasoning for multi-turn evidence gathering.
- * Solved multi-hop retrieval over heterogeneous FHIR resources via graph-based node pruning and information fusion, where LLM-guided schema mappers select relevant fields from unified resource graphs and merge cross-resource evidence into canonical representations for consistent extraction across all resource types.
- * Built a versioned Skill Bank with illustrative markdown skills and executable Python tools, enabling batch training with trajectory analysis and an LLM-based Skill Improver that proposes ADD/REPLACE/DEL skill updates validated against held-out samples via rollback-gated updates to prevent performance regression.

SenseTime Incorporated

Dec. 2020 – Jun. 2021

Algorithm Developer Intern | Mentor: Ruodai Li

Shenzhen, China

• Contactless Vital Signs Estimation from Thermal Imaging using Computer Vision

Python, Signal Processing, Image Processing, Computer Vision

- * Designed and implemented a contactless physiological monitoring algorithm for the **SenseThunder Air** product, integrating **landmark detection** and **homography transformation** to extract regions of interest from thermal video streams for real-time heart rate and body temperature estimation.
- * Applied advanced signal processing methods, including **FFT**, smoothing, and **bandpass filtering**, to recover weak respiratory and cardiac signals from continuous thermal frames.

- * Developed an **end-to-end Python framework** for data preprocessing, algorithm implementation, performance evaluation, and error analysis, enabling robust real-world deployment.

ACADEMIC EXPERIENCE

University of Virginia

Jun. 2022 – Present

Graduate Research Assistant | Ph.D. Advisor: Homa Alemzadeh

Charlottesville, VA

- * **DoC: Learning When to Defer or Commit in Conversational Diagnosis Prediction**

PyTorch, SFT, GRPO, vLLM, DeepSpeed, W&B

- Developed a **self-improving SFT loop** that iteratively regenerates turn-level DEFER/COMMIT pseudo-labels with the updated policy, using only dialogue-level diagnosis labels and requiring no human-annotated commitment times.
- Develop a policy training strategy that applies DEFER-aware supervised fine-tuning initialization followed by trajectory-level reinforcement learning (GRPO), which jointly optimizes commitment timeliness, first-commit accuracy, and post-commit stability, allowing the policy to improve beyond initial supervision.
- Post-trained **Qwen3-1.7B** to outperform larger Qwen3-4B and Doctor-R1-8B baselines, achieving 78.3% and 52.8% first-commit accuracy on two real-world dialogue datasets respectively, while reducing post-commit error by 43% and 24% versus the strongest baseline on each dataset.

- * **EMSDialog: Synthetic Multi-person EMS Conversation Generation from Electronic Patient Care Reports via Multi-LLM Agents**

PyTorch, Multi-LLM Agents, vLLM, LoRA

- Built a scalable, **EHR-grounded multi-agent pipeline** for synthetic EMS dialogue generation, combining iterative **critique-and-refine**, deterministic **rule-based checkers**, and an LLM-based **style checker** to improve clinical factuality, procedural realism, and naturalness.
- Created **EMSDialog**, a dataset of **4,414** annotated EMS conversations with **43 diagnoses**, speaker roles, and topic labels, and showed that **EMSDialog**-augmented training improves conversational diagnosis prediction in **accuracy, timeliness, and stability**.

- * **Expert-Guided Prompting and Retrieval-Augmented Generation for EMS Question Answering**

PyTorch, RAG, FAISS, vLLM, LoRA, DeepSpeed, Wandb, Selenium, BeautifulSoup, Linux, CUDA

- Created **EMSQA**, the first EMS MCQA dataset of 24.3K questions, curated based on public and private sources, covering 10 subject areas and 4 certification levels, and accompanied by a structured, subject area aligned EMS knowledge base (KB) with 40K documents and 4M real-world patient care reports.
- Developed two techniques to inject domain expertise into Large Language Models: 1) **an expertise-guided prompting (Expert-CoT)** that encourages step-by-step reasoning from a domain-specific perspective. 2) **an expertise-guided RAG (ExpertRAG)** that retrieves expertise-aligned knowledge from curated EMS KBs and patient records.
- Benchmarked multiple LLMs on EMSQA, evaluating performance across certification levels and subject areas, and compare our framework against SOTA RAG methods. Experimental results show that combining Expert-CoT and ExpertRAG yields up to a **4.67% improvement in accuracy**. Notably, the **32B expertise-augmented models pass all the EMS certification** simulation exams.

- * **DKEC: Domain Knowledge Enhanced Multi-Label Classification for Diagnosis Prediction**

PyTorch, UMLS, BeautifulSoup, Prompting, BERT, GNN, GPT, FSDP, Wandb, Linux, NER

- Automated **heterogeneous knowledge graph** construction from 3,000+ medical webpages (Wikipedia, MayoClinic, MedlinePlus) using BeautifulSoup, Web APIs, and LLM prompting (chain-of-thought), extracting 5,000+ normalized medical entities via UMLS API.
- Designed a **label-wise attention** mechanism that incorporates heterogeneous knowledge graphs to language models, to improve multi-label classification by **5% in micro f1** compared with SOTAs on MIMIC-III datasets.

- * **EgoEMS: A High-Fidelity Multimodal Egocentric Dataset for Cognitive Assistance in EMS**

C++, Python, Android, Vision-Language Model, ASR models, Action Recognition

- Designed and executed benchmarking pipeline for **zero-shot audio models** (Whisper, Whisper-Timestamped, Google Speech, Gemini-2.5-Pro), evaluating ASR accuracy, latency, and word-level timestamp precision in self-collected real-world EgoEMS dataset.
- Conducted comprehensive comparative analysis of **zero-shot vision-language models** (Qwen-2.5-VLM, VideoLLaMA-3, Gemini-2.5-Pro) for **video understanding**, measuring **action classification** accuracy,

temporal segmentation quality, and inference efficiency to guide model selection for downstream applications.

* **Real-Time Multimodal Cognitive Assistant for Emergency Medical Services**

Python, PyTorch, (EMSTiny)BERT, GNN, CUDA, NVIDIA Jetson

- Cleaned EMS electronic patient care reports (ePCRs) with regex-based normalization, schema mapping, and quality filters to create train/val/test splits.
- Developed **EMSTinyBERT** (15M params), fine-tuned on 4K ePCRs plus curated guideline text; proposed *group-wise* training to address rare classes and boost minority-label recall.
- Deployed the model on an NVIDIA Jetson (CUDA) as an on-device inference pipeline; achieved **80.0%** accuracy with **0.31 s** end-to-end latency.

Chongqing University

Jul. 2019 – Nov. 2019

Undergraduate Research Assistant | Mentor: Juan Yu

Chongqing, China

* **Intelligent Diagnosis of Alzheimer's disease Based on Computer Vision**

Python, CNN, Activity Recognition, Object Detection

- Designed and implemented an AI-based system to automatically assess the severity of Alzheimer's disease using video and image data, enabling objective and scalable cognitive evaluation.
- Developed human eye state recognition GUI using **Dlib** for automated cognitive assessment.
- Designed video analysis pipeline integrating **OpenPose-based posture estimation, image morphology processing**, and **Fast R-CNN** detection to detect patient's action for Alzheimer's severity evaluation.
- Trained a **CNN with a ResNet-50 backbone** in PyTorch to classify patient hand-drawn geometric figures, improving diagnostic accuracy and enabling early detection of cognitive impairment.

SERVICE

• **Teaching Assistant:**

ECE Statistical Machine Learning (Fall 2022);
Dependable Computing System (Fall 2025).

• **Reviewer:**

2024: ICRA;
2025: NAACL; ACL; IJCAI; AAAI; NeurIPS; ACM Transactions on Computing for Healthcare.
2026: AAAI; ACM Transactions on Computing for Healthcare; ICLR; IJCAI; NeurIPS; COLM.
2027: AAAI .

• **Volunteer:**

2024: EMNLP; LinkLab Open House;
2025: Link Lab FlashTalk; Link Lab Research Day;
2026: AAAI; LinkLab Open House;

• **Mentoring:**

Saahith Janapati; Shruti Bala; Sion Kim; Abhishek Satpathy; Sahil Murtaza

COURSE PROJECTS

N-Version Programming on LLMs | *GPT-3.5, PaLM-1, Llama-2, Ensembling, vLLM*

Aug. 2023 – Dec. 2023

- * Improved multiple-choice question answering accuracy by merging responses from multiple LLMs.
- * Designed majority and weighted-majority voters over GPT-3.5/PaLM-1/Llama-2 outputs, yielding **+5%** accuracy.

BERT Visualization and Interpretation | *PyTorch, BERT, SNLI*

Aug. 2022 – Dec. 2022

- * Analyzed whether BERT produces reasonable layer-wise embeddings.
- * Visualized anisotropy by sampling words and computing cosine similarities across 12 layers.
- * Studied bias by removing “female”-related sentences from SNLI and measuring effects.
- * Investigated redundancy by freezing individual layers on classification tasks.

Phishing Website Detection | *Python, SVM, RF, GBDT, Java, PhishTank API*

Jan. 2021 – May 2021

- * Designed phishing detection algorithms and a phishing query web application.

- * Engineered URL features (domain, IP, DNS, redirects, protocol, etc.) for model inputs. And trained SVM/Random Forest/GBDT and built an ensemble to boost performance; Build an en-to-end pipeline by integrating trained ensembled model with PhishTank API.